

Adnan Mashrur Sadad

+60 11-3968 7435 | adnanmashrursadad@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Software Engineering graduate (UTM Malaysia) building production-grade AI/ML and GenAI systems — RAG pipelines, LLM evaluation loops, and deployed ML services (FastAPI, Docker, CI/CD). Shipped four end-to-end ML/AI products from data pipeline to deployed API, achieving up to 0.98 ROC-AUC on real-world classification tasks.

Technical Skills

- **GenAI / LLM Systems:** RAG, Agentic Workflows, Text2SQL, Prompt Engineering, Structured Output Validation, LLM Evaluation & Repair Loops, Groq API, Whisper
- **Machine Learning & Data:** XGBoost, Scikit-learn, TensorFlow, Pandas, NumPy, Computer Vision (CNN/ANN), OCR, Predictive Analytics, Monte Carlo Simulation, Hugging Face
- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, SQL, Dart
- **Frameworks & Tools:** FastAPI, React, SQLAlchemy, Streamlit, Recharts, Spring, Flutter, Git
- **Deployment & Architecture:** Docker, REST APIs, Microservices, CI/CD (GitHub Actions), OpenAPI
- **Core CS:** Data Structures & Algorithms, OOP, Operating Systems, Databases, Software Engineering

Projects

InterviewPilot – AI-Powered Mock Interview Platform

2026 – GitHub

Python, FastAPI, SQLAlchemy, React + TypeScript, Groq API (Whisper-large-v3, Llama 3.3 70B), Docker, GitHub Actions

- Architected a full-stack mock interview platform (FastAPI/SQLAlchemy backend, React + TypeScript frontend) using Whisper-large-v3 for speech transcription and Llama 3.3 70B for automated interview evaluation
- Built a rubric-based evaluator that returns schema-validated JSON (technical accuracy, clarity, depth) with a self-correcting repair loop that re-prompts the model on its own invalid output
- Designed an agentic follow-up question generator that reasons over candidate answers to decide when to probe deeper, mirroring real interviewer behavior
- Enforced engineering rigor with a pytest suite (17 tests, 87% backend coverage, fully mocked Groq calls), Dockerized deployment, and GitHub Actions CI on every push

AI Financial Fraud Detection System

2025

Python, XGBoost, Scikit-learn, FastAPI, Streamlit

- Engineered an ensemble ML pipeline (Random Forest, XGBoost, Isolation Forest) on the public Kaggle credit-card-fraud dataset (284,807 transactions, 0.17% fraud rate), achieving 0.98 ROC-AUC and 0.89 PR-AUC on held-out data
- Addressed extreme class imbalance with SMOTE and class-weighted loss, prioritizing PR-AUC as the primary evaluation metric for rigorous minority-class assessment
- Deployed a modular inference service with FastAPI, Pydantic validation, and OpenAPI docs, paired with a Streamlit dashboard for batch scanning and model explainability

AI Chatbot Engineer Assessment — Mindhive Asia

2025 – GitHub

FastAPI, RAG, Text2SQL

- Designed a multi-turn conversational AI with stateful memory and intent-based agentic planning, sustaining coherent context across 3-5 sequential user turns
- Built FastAPI microservices including a RAG-powered product knowledge endpoint and a Text2SQL outlet-query endpoint, validated against 200+ product documents with SQL-injection protection
- Evaluated retrieval quality with a hand-labeled 20-query harness against the live product/outlet corpus: 85% Hit Rate@3 overall (100% on outlet lookups), MRR 0.86
- Delivered a complete repository with OpenAPI specification, test suite, architecture diagrams, and a hosted live demo

WC26 Predictor – World Cup 2026 Forecasting Dashboard

2026 – GitHub

Python, XGBoost, FastAPI, React, TypeScript, Recharts

- Built an end-to-end football analytics pipeline using historical match results, FIFA rankings, rolling form, squad proxies, and head-to-head features to model match outcomes and expected goals
- Benchmarked outcome models on held-out matches: XGBoost reached 59.4% accuracy (vs. 47.8% majority-class baseline) with a calibrated 0.522 Brier score, and a Poisson goals regressor achieved 1.02 MAE
- Ran Monte Carlo simulations over the calibrated probabilities to project group qualification, knockout paths, and championship odds, surfaced in an interactive dashboard with tournament-path curves and a predicted-vs-actual audit view

Work Experience

Joget Inc.

Oct 2024 – Mar 2025

Software Developer Intern (Hybrid)

- Built and deployed enterprise workflow automations and UI components on the Joget DX low-code platform, digitizing core business processes for clients across multiple departments
- Integrated REST APIs with external services, streamlining data flow and reducing manual processing effort by 30%
- Shipped production features within Agile sprints alongside senior engineers, QA, and product managers, covering the full SDLC from design through deployment
- Deployed and maintained workflow modules through CI/CD pipelines, contributing to recurring production release cycles for enterprise clients

Education

Universiti Teknologi Malaysia (UTM), MJIIT

2021 – 2026

Bachelor of Software Engineering, CGPA: 3.43/4.00 — Dean's List 2022/2023

Kuala Lumpur, Malaysia

- Undergraduate thesis: OCR-based personal finance tracking system for automated income-tax-ready expense classification
- Relevant coursework: Artificial Intelligence & Machine Learning, Software Engineering, Data Structures & Algorithms, Operating Systems, Computer Architecture

Maple Leaf International School

A Levels & O Levels

Dhaka, Bangladesh

Achievements & Leadership

- Gold Medal & Best Video Award, myHCI-UX Student Design Challenge (Oct 2025) — national university competition for UX design and innovation
- Top Performer, Computational Intelligence course, UTM (2023)
- Secretary Treasurer, SOFEA Society (2022–2023); organized AI workshops for 200+ students